

Lucas Picard

B.S./M.S. Mechanical + B.S. Robotics Engineering | Mechatronics · CAD · Embedded Systems · Prototyping

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SUMMARY

I am a mechanical engineering student at WPI (3.75 GPA, Dean's List) on a combined B.S./M.S. in Mechanical Engineering with a second major in Robotics Engineering, captivated by the challenge of building things others think are impossible. I design and bring real hardware to life — embedded electronics, robotic systems, and CAD assemblies — and direct AI to write the software and automation that drives them. This past summer I automated digital engineering processes as an engineering intern at a contract electronics manufacturer, and as a dual US/German citizen, bilingual in English and German, I am driven to push the boundaries of what is possible in the STEM field.

EXPERIENCE

Microboard Processing — *Engineering Intern, Document & Controls* Jun – Aug 2026

Seymour, CT — contract electronics manufacturer (ITAR-regulated).

- Automated digital engineering and documentation processes alongside an industrial engineer, streamlining manufacturing workflows.
- Built standardized SMT/PCB assembly process-flow documentation that sharpened consistency and clarity across the production line.
- Applied disciplined data-handling and documentation practices within a regulated, compliance-sensitive environment.

Picard Studio — *Studio Assistant & Operations* Feb 2020 – Present

Southbury, CT — art education business (online courses & membership).

- Built and maintained KPI dashboards tracking MRR, ARPU, retention, churn, and acquisition to drive business decisions.
- Automated monthly metrics reporting through the Kajabi API and GitHub Actions, piping data straight into Google Sheets.
- Authored a membership cancellation analysis (13 charts, 15-page report) that surfaced renewal-window and engagement-based retention signals.
- Delivered freelance CAD work — designing, prototyping, and invoicing a wall-mount equipment cradle.

Aquila's Nest Vineyards — *Service Staff (Seasonal)* Jul – Nov 2024

New Morning Market — *Grocery Clerk* Aug – Nov 2023

EDUCATION

Worcester Polytechnic Institute — *B.S./M.S. Mechanical Engineering + B.S. Robotics Engineering*

Expected 2029

Worcester, MA — Entered with 21 AP transfer credits (Calculus, Physics C Mechanics & E&M, Computer Science A, Studio Art, and US Government). GPA: 3.75 / 4.00. Dean's List.

Completed coursework: Statics, Dynamics, Thermodynamics, Manufacturing Science & CNC Machining, Introduction to Robotics, Intro to ECE, Calculus III–IV, Differential Equations, Linear Algebra.

Upcoming: Unified Robotics I–IV, Fluid Mechanics, Heat Transfer, Stress Analysis, Control Engineering, Materials Science, Software Engineering.

Activities: Combat Robotics, Club Badminton, Intramural Soccer.

Pomperaug High School — *Southbury, CT* 2021 – 2025

GPA: 4.06 / 4.5 scale. Honor Roll, all years. German Seal of Biliteracy.

ENGINEERING PROJECTS

SCARA Robot Arm — *Personal mechatronics build* 2026 – Present

- Turned a salvaged \$150 Anet A8 3D printer into a teleoperated 4-axis SCARA arm for about \$120 in new parts, with custom 3D-printed arm segments and GT2 pulleys up to 100 teeth.
- An Xbox controller drives the end effector through Cartesian space at 25 Hz, with a Python host solving the inverse kinematics live and sending G-code to Marlin.
- Integrated AS5600 absolute magnetic encoders on a three-bus I²C design for switch-free homing that survives power cycles; base, elbow, and Z are live, with wrist and gripper next.

PenGPT AI Smart Pen — *Embedded hardware / PCB design*

Mar 2026 – Present

- Co-designing the hardware with a teammate for a pen that turns handwriting on ordinary paper into text, tracking its own motion with two IMUs and a magnetometer around a Sophgo SG2002 Linux processor.
- Verified the 268-part KiCad schematic pin by pin against manufacturer datasheets, catching three faults ERC missed; the 4-layer bring-up board is placed and routing is next.

CT Used-Car Scraper — *Python automation tool*

2026

- Directed AI to build a Python + Playwright system that scrapes 36 Connecticut dealer sites every 6 hours on GitHub Actions for \$0/month, tracks price history in SQLite, scores deals, and sends Discord alerts; it found the car I bought.

FRC Robotics — *Team Captain & Lead Driver · 1000+ hours*

2021 – 2025

- Led CAD, mechanical build, and electrical integration across subteams, designing robot subsystems in Onshape / SolidWorks and troubleshooting rapidly in the pit.
- As primary driver, earned the team's first New England District Championship qualification in 10 years and its first CT State Championship (off-season).

TECHNICAL SKILLS

CAD & Design: SolidWorks, Onshape, GD&T, design for manufacturability

Electronics: KiCad PCB design, soldering, embedded systems (ESP32 / Arduino), closed-loop control, sensors & encoders

Prototyping: 3D printing, manual milling, laser cutting, CAM

Software & Data: Python, Playwright, REST API automation, GitHub Actions, Excel / Google Sheets KPI analysis

Other: Technical documentation, ITAR-aware data handling

AWARDS · LANGUAGES · INTERESTS

Awards: Dean's List (WPI, Fall 2025); Honor Roll (Pomperaug, all years).

Languages: English (native), German (Seal of Biliteracy) — dual US/German citizen.

Interests: Acrylic painting, drawing, music performance, robotics.